

GeoMatch: photo geolocation as visual retrieval

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1 Approach

Predict a photo’s GPS coordinate (12 European countries, 11,758 labelled images, 2,400 held out) with *one* $\leq 5,000,000$ -parameter model, no pretrained weights, no external data.

We predict by **retrieval**: embed query and bank, copy the nearest match’s coordinate. Retrieval is not continuous — its output is restricted to the 11,758 training coordinates — but that discretisation is cheap: the geographically nearest training photo is a median 3.1 km away and within 50 km for 99.8% of queries, far below the error we achieve, whereas a geo-cell classifier is bounded by its cell size. The gain is that the encoder never regresses a coordinate; it only learns which photos were taken near each other, which supervises densely from every pair in a batch.

2 Model (Fig. 1)

One RegNetY-400MF trunk [1] (**weights=None**) is shared across three views: the full frame at 512^2 and aligned left/right crops at 256^2 , which keep detail a single down-scaled frame destroys.

Global branch. Multi-scale GeM pooling [2] over stages 2–4, fused across views, through an identity-initialised 384×384 layer, L2-normalised to a descriptor g . **Local branch.** Stage-4 features average-pooled to 4×4 per view (48 tokens), projected $440 \rightarrow 64$, L2-normalised to tokens T .

The shortlist is the cosine top-200 of $\langle g_q, g_b \rangle$, rescored by a symmetric MaxSim late interaction [3]:

$$s(q, b) = 0.2 \langle g_q, g_b \rangle + 0.8 \cdot \frac{1}{2} \left(\overline{\max}_j T_q T_b^\top + \overline{\max}_i T_q T_b^\top \right)$$

The top-1 candidate’s coordinate is the prediction.

3 Training

Evaluated on the official 5-fold split: train on four folds, predict the fifth, fixed epoch count, no checkpoint selection.

(1) **BYOL pretraining** [4] of the trunk, labels unused, 160 epochs. (2) **Country-aware retrieval fine-tuning**, 40 epochs, AdamW (8×10^{-5} trunk, 6×10^{-4} heads), physical batch 32 with 2 gradient-accumulation steps (effective 64), EMA 0.999. Each group is an anchor, two geographic positives and one *same-country* hard negative 60–700 km away, re-mined from the current EMA encoder every epoch. Positives are the 8 most descriptor-similar of the 64 geographically nearest that lie within 50 km; for the 10.7% of anchors with fewer than 8 such neighbours the sampler falls back to the 8 nearest *regardless of distance*, so “ ≤ 50 km” holds for $\approx 89\%$ of anchors, not all. Anchors from the weakest countries are oversampled (DE 3.0 $\times$, FR 2.8, PL 2.0, IT/ES 1.7, SE 1.4, GB

1.3). Objective: 0.65 retrieval + 0.2 classification regulariser.

4 Results

pooled 5-fold OOF	runs	median (km)	≤ 50 km
From-scratch retrieval baseline	1	89.2	45.3%
+ BYOL + country-aware, 22 ep	1	57.7	48.8%
+ 40 epochs (shipped recipe)	3	55.6 ± 1.1	48.8%
(non-compliant) 3-model ensemble	1	52.0	49.7%
(non-compliant) 6-model ensemble	1	49.7	50.1%

Table 1: Shipped-recipe row is mean $\pm$ sample s.d. of three independently seeded 5-fold runs (55.60 / 57.12 / 57.71 km; the ≤ 50 km column is likewise the 3-run mean). **Best observed is 55.60 km**; the submitted model uses the configuration’s default seed, which was not among the three, so no seed was chosen on evaluation results. Ensembles are listed to record that the gain is real and was rejected — the rules require a single ≤ 5 M model.

Which half did the work? The -31 km row changes two things at once, so we separate them on fold 0: baseline (no BYOL, no country-aware, ep 6) 76.1 km; country-aware fine-tuning alone 70.0 km from baseline weights (ep 12) and 71.9 from a joint start (ep 24); adding the BYOL initialisation **53.6** (ep 22). Country-aware sampling buys ~ 6 km, the BYOL trunk ~ 16 km more — both contribute, BYOL more. Single fold and unequal epoch counts, so indicative rather than decisive.

Extending 22 to 40 epochs is within the ± 1.1 km seed spread, so we do not claim it as an improvement; 40 was kept because the schedule was fixed in advance.

5 Where the error comes from

Splitting each query into *located* (≤ 50 km), *mis-ranked* (a ≤ 50 km bank image was in the top-200 but not returned first) and *never retrieved* separates the two failure modes.

	located	mis-ranked	never retrieved
Pooled	49.1%	30.3%	20.7%
Iceland	82.9%	15.2%	1.9%
France	25.2%	32.7%	42.1%
Germany	14.2%	40.6%	45.2%

Table 2: Computed on one seed (55.60 km run), so the pooled “located” figure is 49.1% rather than the 3-run mean of 48.8% in Table 1. Ranking is the larger loss pooled, but in Germany and France $\sim 45\%$ of queries retrieve nothing nearby at all: both stages fail, and a better reranker alone could not have closed it.

Germany and France dominate what is left. They are 17% of the evaluation set; excluding them the pooled median falls from 55.6 to **31.7** km, and merely bringing them to Poland’s level (the next worst) would give 41.8 km. They do not account for *all* of the gap, but no other single factor comes close.

GeoMatch: Single-Model Visual Retrieval Architecture

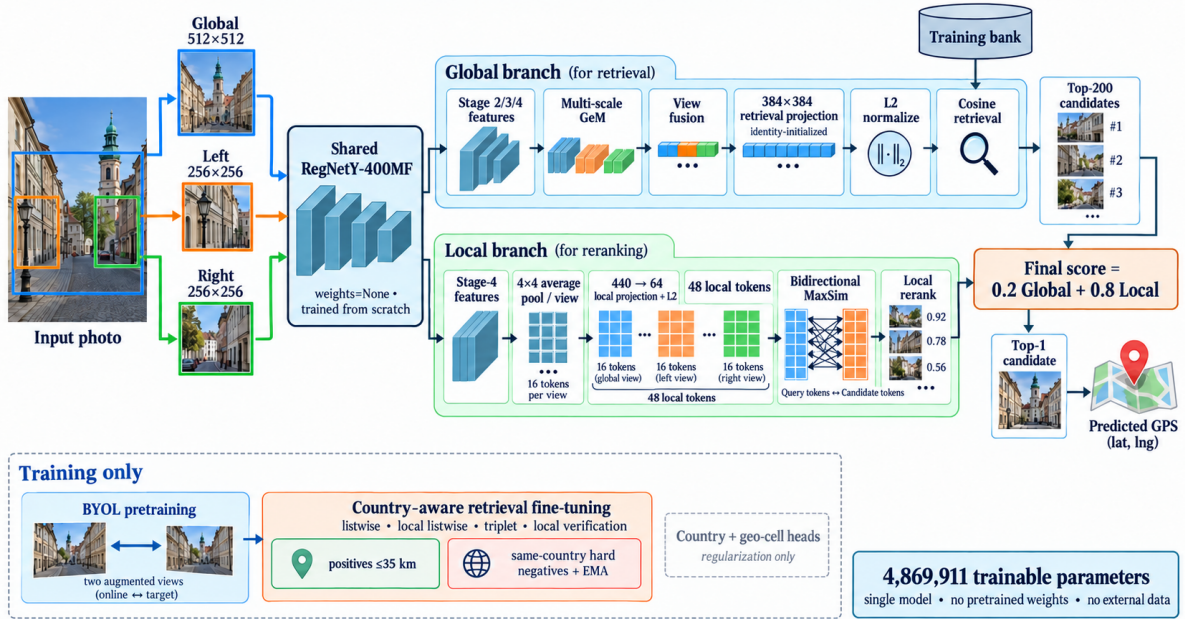


Figure 1: One from-scratch RegNetY-400MF trunk, shared across a global view and two crops, feeds a global-descriptor retrieval branch and a local-token late-interaction reranker. Classification heads regularise training and are unused at inference. 4,869,911 parameters. (The diagram’s “positives ≤ 35 km” label is stale: the shipped config uses 50 km, see §3.)

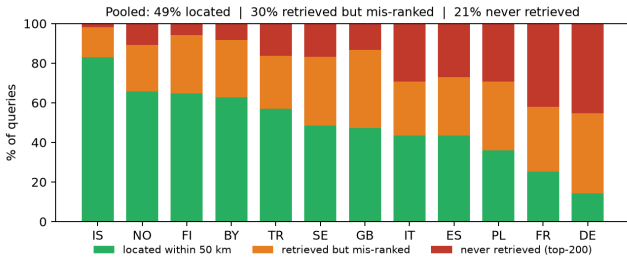


Figure 2: Error decomposition per country (same single seed as Table 2), sorted by success rate.

6 What did not work

Decode-time repairs on frozen descriptors (5-fold evidence). A learned candidate reranker was tried four times (linear, residual, pairwise, and one gated on a held-out inner fold) and degraded the crossfit every time; the inner-fold gate passed only because the probe fold was held out from the reranker but not from the *encoder*. Kernel-density consensus decoding gave 72 km against 57.7. Gating the shortlist on predicted country cost 4 km pooled (country accuracy is 68%, and a wrong gate is unrecoverable). Decoding from the auxiliary geo-cell head instead of retrieval gave DE 490 / FR 523 km against retrieval’s 534 / 551: two decoders reading the same descriptor fail equally, which locates the problem upstream of the decoder — though it does not establish what the descriptor is missing.

Encoder variants (fold 0 only — weaker evidence). Adding a random-resized-crop to the global view was worse at every matched epoch (63.7 vs 58.1 km at epoch 32), consistent enough to call a backfire; it discards the horizon and scene layout. Raising the local grid from 4×4 to 8×8 was *not* worse (58.3 vs 60.7 km at epoch 40) — we kept 4×4 as the simpler option, not because the variant failed; extending it to 48 epochs drifted to

61.8 km. Interpret both cautiously: the shipped recipe’s own fold-0 median varies by 5.3 km across epochs 20–40, so single-fold gaps of that size are not decisive.

7 Limitations

Germany and France are the dominant failure and we did not solve them. The evidence locates the problem in the representation rather than the decoder, but we never ran the experiment that would say *why* — e.g. probing whether the descriptor carries any within-country position signal. Country-aware oversampling targeted exactly this and moved every other weak country but not these two. Also: the failed-variant evidence is single-fold and confounded by epoch count, and the figures are rebuilt from cached out-of-fold predictions that are not in the repository, so they cannot be regenerated from a clean clone.

8 Submission

The recipe retrained once on all 11,758 images with no held-out fold, from an existing BYOL trunk, using the configuration’s default seed: 4,869,911 parameters, **67 min of fine-tuning** on one RTX 4000 Ada (excluding BYOL pretraining, ≈ 2 h 40 min, which was reused rather than re-run), plus ≈ 5 min to write `predictions.csv`.

- [1] Radosavovic et al. Designing Network Design Spaces. CVPR 2020.
- [2] Radenović, Tolias, Chum. Fine-tuning CNN Image Retrieval with No Human Annotation. TPAMI 2019.
- [3] Khattab, Zaharia. ColBERT: Late Interaction over BERT. SIGIR 2020.
- [4] Grill et al. Bootstrap Your Own Latent. NeurIPS 2020.